Cours MALG & MOVEX

Vérification d'une annotation

Dominique Méry
Telecom Nancy, Université de Lorraine

Année universitaire 2025-2026 (23 mars 2026 9:52 P.M.)

Plan

1. WP calculus in Frama-c
2. First annotation
3. Second annotation

Sommaire

1. WP calculus in Frama-c
2. First annotation
3. Second annotation

Current Summary

1. WP calculus in Frama-c
2. First annotation
3. Second annotation

Verification Condition for Assignment

Listing 1 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

Verification Condition for Assignment

Listing 2 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$

Verification Condition for Assignment

Listing 3 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$

Verification Condition for Assignment

Listing 4 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)

Verification Condition for Assignment

Listing 5 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 6 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 7 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 8 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \vdash Q(x)$

Verification Condition for Assignment

Listing 9 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \vdash Q(x)$

  Assume $\{$
  $P(x1)$
- $x = e(x1)$
  $\}$
  Prove:$Q(x)$
-

Current Summary

1. WP calculus in Frama-c
2. First annotation
3. Second annotation

Annotation simple (I)

Listing 10 – an1.c

```
int main(void){
  signed long int x,y,z; //     int x,y,z;
  x = 1;
  /*@  assert x == 1; */
  y = 2;
  /*@ assert x == 1 && y == 2; */
  z = x *y;
  /*@  assert x == 1 && y == 1 && z == 2; */
  return 0;
  }
```

Annotation simple(I)

```
[kernel] Parsing an1.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 2 goals scheduled
[wp] Proved goals:    4 / 4
  Terminating:     1
  Unreachable:     1
  Qed:             2
```

Annotation simple (I)

```
Goal Assertion 'l1' (file an1.c, line 3):
Assume {
  Type: is_sint32(x) /\ is_sint32(y).
  (* Initializer *)
  Init: x = 12.
  (* Initializer *)
  Init: y = 24.
}
Prove: (2 * x) = y.
Prover Qed returns Valid
```

Annotation simple (I)

```
------------------------------------------------------------

Goal Assertion 'l2' (file an1.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(y).
  (* Initializer *)
  Init: x_1 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_1) = y.
  Have: (1 + x_1) = x.
}
Prove: (2 + y) = (2 * x).
Prover Qed returns Valid
```

Annotation simple(I)

```
[kernel] Parsing an1.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 2 goals scheduled
[wp] Proved goals:    4 / 4
  Terminating:     1
  Unreachable:     1
  Qed:             2
```

Current Summary

1. WP calculus in Frama-c
2. First annotation
3. Second annotation

Annotation simple (II)

Listing 11 – an2.c

```
void ex(void) {
  int x=12,y=24;
  //@ assert l1: 2*x == y;
  x = x+1;
  //@ assert l2:  y == 2*(x-1);
  x = x+2;
  //@ assert l3:  y+6 == 2*x;
}
```

Annotation simple (II)

```
[kernel] Parsing an2.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 3 goals scheduled
[wp] Proved goals:    5 / 5
  Terminating:     1
  Unreachable:     1
  Qed:             3
```

Annotation simple (ii)

```
Goal Assertion 'l1' (file an2.c, line 3):
Assume {
  Type: is_sint32(x) /\ is_sint32(y).
  (* Initializer *)
  Init: x = 12.
  (* Initializer *)
  Init: y = 24.
}
Prove: (2 * x) = y.
Prover Qed returns Valid
```

Annotation simple (ii)

```
Goal Assertion 'l2' (file an2.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(y).
  (* Initializer *)
  Init: x_1 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_1) = y.
  Have: (1 + x_1) = x.
}
Prove: (2 + y) = (2 * x).
Prover Qed returns Valid
```

Annotation simple (ii)

```
Goal Assertion 'l3' (file an2.c, line 7):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(x_2) /\ is_s
  (* Initializer *)
  Init: x_2 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_2) = y.
  Have: (1 + x_2) = x_1.
  (* Assertion 'l2' *)
  Have: (2 + y) = (2 * x_1).
  Have: (2 + x_1) = x.
}
Prove: (6 + y) = (2 * x).
Prover Qed returns Valid
```

Listing 12 – an2bis.c

```
void ex(void) {
  int x=12,y=24;
  //@ assert l1: 2*x == y;
  x = x+1;
  //@ assert l2:  y == 2*(x-1);
   x = x+2;
  //@ assert l3:  y+6 == 2*x;
}
```

¡